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User Activity Monitoring Tool

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1. Introduction

This project is a Windows-based User Activity Monitoring Tool developed using C++ and the Windows API. The system monitors all running processes on a computer and tracks their activity, including when a process starts, when it ends, and how long it runs. It provides a graphical interface along with file logging for easy tracking and record keeping.

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2. Objectives

The main objectives of this project are:

- To monitor user activity through system processes
- To detect process start and stop events
- To calculate execution duration of processes
- To display logs in a graphical interface
- To store logs in a file for future use

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3. Problem Statement

Users are often unaware of background processes running on their system. It becomes difficult to track which applications are active and how long they are used. This project solves the problem by automatically monitoring and recording user activity through process tracking.

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4. Features of the System

- Real time monitoring of processes
- Detection of process start
- Detection of process termination
- Duration calculation of each process
- Graphical User Interface (GUI)
- Start and Stop monitoring buttons
- Auto scrolling log display
- File logging in **activity_log.txt**

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5. Tools and Technologies Used

- **Programming Language:** C++
- **Platform:** Windows
- **API:** Windows API (Win32)

Libraries Used:

- windows.h
- tlhelp32.h
- string
- set
- map
- fstream
- ctime

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6. Working Methodology

1. The system captures all running processes using Windows API snapshot.
2. It stores process IDs and compares them after every 2 seconds.
3. If a new process appears, it logs **PROCESS STARTED** with time.
4. If a process ends, it logs **PROCESS ENDED** with duration.
5. Duration is calculated using system time functions.
6. Logs are shown in GUI and saved into a text file.

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7. Output Description

GUI Output:

- PROCESS STARTED: chrome.exe | time
- PROCESS ENDED: chrome.exe | time (Duration: X sec)

File Output:

All logs are saved in:
activity_log.txt

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8. Advantages

- Easy to use interface

- Real time activity tracking
- Automatic logging
- Useful for monitoring user behavior
- Helps in system management

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9. Limitations

- Only works on Windows OS
- Checks processes after fixed time interval
- Does not track CPU or memory usage

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10. Future Improvements

- Add CPU and RAM usage tracking
- Export logs to PDF or Excel
- Add process filtering and search
- Improve GUI design
- Add alert system

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11. Conclusion

This project successfully implements a User Activity Monitoring Tool using C++ and Windows API. It tracks process activity efficiently by recording start time, end time, and duration. The system is simple, useful, and can be enhanced further for advanced monitoring features.